



Blueprint for Applied Physics
Department: F.E. (All Branches)
Internal Assessment Test–I (A.Y. 2026–27)
NEP–2020

Subject: Applied Physics

Semester: I

Max. Marks: 15

Duration: 1 Hour

Note: All questions are compulsory.

Q. No.	Questions	Marks	BL	CO
Q.1	Attempt any two of the following:			
(a)	Theory and formula of Anti-reflection coating/ Explanation of Newton's interference Rings.	3	L2	CO3
(b)	Numerical on determination of wavelength and diameter of Newton's interference Rings.	3	L3	CO3
(c)	Interference in thin film of uniform thickness and the explanation of maxima, minima conditions.	3	L2	CO3
Q.2	Attempt any one of the following:			
(a)	Theory and expression of numerical aperture (N.A.) / Types of losses in optical fibers.	4	L3	CO2
(b)	Theory of various types of optical fibers and numerical problem on acceptance angle and N.A.	4	L3	CO2
Q.3	Attempt any one of the following:			
(a)	Theory of He-Ne laser and resonant cavity.	5	L2, L3	CO1
(b)	Explanation of the terms like; absorption, spontaneous, stimulated emission, pumping, metastable state, population inversion. Theory and formula of coherence length and coherence time.	5	L2, L3	CO1

BL: Bloom's Taxonomy Levels

(L1: Remembering, L2: Understanding, L3: Applying, L4: Analyzing, L5: Evaluating, L6: Creating)

CO: Course Outcomes